

Shplemini Sparse Masking

This note proves the KZG masking lemma used to replace the full-size random `gemini_masking_poly` in the Gemini + Shplonk + KZG path of Shplemini by a sparse polynomial with only $2d$ random coefficients.

We first prove the statement over a rational function field. A concrete Fiat-Shamir transcript is obtained by specialising the indeterminates; the proof is valid at every specialisation outside the explicitly defined exceptional locus.

Setup

Let

$$N = 2^d, \quad d \geq 4, \quad 2d < N.$$

Let $\mathbb{F}$ be the scalar field and write $E_j(X) = X^j$ for the monomial basis of $\mathbb{F}[X]_{<N}$. Let e be the maximum end-index of the masked polynomial data. Choose the smallest even integer E such that

$$e \leq E \leq N, \quad \{E-1, E-2\} \cap \{N/2^\ell, N/2^\ell - 1\} = \emptyset \quad (1 \leq \ell < d).$$

Equivalently, if $B = N/2^\ell$ is dyadic, then $E \notin \{B, B+2\}$. In the main argument we assume

$$N/2 < E \leq N.$$

When $E \leq N/2$, the dense polynomial space is smaller and the low-tail containment argument applies after deleting the unused dense columns. The saturated case $d = 3$ is recorded separately in Appendix B.

Define the tail-halving support

$$S = [E-1, E-2, N/2, N/2-1, N/4, N/4-1, \dots, 2, 1],$$

truncated to exactly $2d$ entries. The non-exceptional choice of E makes these entries distinct. The prover samples

$$M(X) = \sum_{s \in S} c_s X^s, \quad c_s \xleftarrow{\$} \mathbb{F}$$

with the c_s independent.

Work over

$$K = \mathbb{F}(u_0, \dots, u_{d-1}, r_0, \dots, r_{d-1}, \tau),$$

where the u_i are Gemini folding challenges, the r_i are Gemini opening points, and τ is the KZG trapdoor. Shplonk batching challenges and the final KZG opening point are used only in the reduction from the full transcript to the Gemini leakage map.

For this support, with

$$A_k = \{s \in S : \lfloor s/2^k \rfloor \geq 1\}, \quad C_k = A_k \setminus A_{k+1} \quad (0 \leq k \leq d-2), \quad C_{d-1} = A_{d-1},$$

the fresh-column blocks are

$$C_0 = \{1\}, \quad C_k = \{2^{k+1} - 1, 2^k\} \quad (1 \leq k \leq d-2), \quad C_{d-1} = \{E-1, E-2, N/2\}.$$

Indeed, the dyadic pair $\{2^{k+1} - 1, 2^k\}$ is exactly the part of the fixed halving tail with

$$2^k \leq s < 2^{k+1},$$

so it lies in $A_k \setminus A_{k+1}$. The column 1 is the unique support point with $1 \leq s < 2$, giving C_0 . Since $E > N/2$, the top pair $E-1, E-2$ lies in A_{d-1} ; the non-exceptional condition $E \notin \{B, B+2\}$ for dyadic B ensures it is disjoint from every dyadic halving pair. Finally $N/2 \in A_{d-1}$, while $N/2 - 1 \notin A_{d-1}$, so the only columns surviving to level $d-1$ are $E-1, E-2, N/2$.

Exceptional locus

Let R_E denote the active rank witness:

$$R_E = \begin{cases} \det B_E, & E > 3N/4, \\ \Delta_E^{\text{lo}}, & N/2 < E \leq 3N/4. \end{cases}$$

The factorisations of $\det B_E$ and Δ_E^{lo} are given in Sections 7 and 8. The exceptional locus is the zero locus of

$$\mathcal{E}_E = R_E \cdot \prod_{t=0}^{d-1} (\tau + r_t) \cdot D_{\text{Shplonk/KZG}},$$

where $D_{\text{Shplonk/KZG}}$ is the product of the Shplonk and KZG denominators that occur in the transcript reduction: the factors $\tau - z_i$, $z - z_i$, $\tau - z$, and the pairwise differences needed for the batched opening points to be distinct.

All Lagrange and affine factors needed for the rank argument already occur as factors of R_E . Thus they are part of the explicit rank-witness zero locus, not an additional informal bad set. After specialising to a finite field, Schwartz-Zippel bounds the failure probability by $\deg(\mathcal{E}_E)/|\mathbb{F}|$.

Theorem

Let

$$V_E = \text{span}\{X^s : 0 \leq s < E\}, \quad V_S = \text{span}\{X^s : s \in S\}.$$

For $P \in V_E$, let P_t be the polynomial obtained after t Gemini folds, and define the Gemini leakage map

$$G_E : V_E \rightarrow K^{2d}, \quad G_E(P) = (P_0(\tau), P_0(-r_0), \dots, P_{d-1}(\tau), P_{d-1}(-r_{d-1})).$$

Let $G_S = G_E|_{V_S}$. Outside the exceptional locus $\mathcal{E}_E = 0$,

$$\text{im } G_E \subseteq \text{im } G_S.$$

More precisely:

$$\begin{aligned} E > 3N/4 : & \quad \text{rank } G_S = 2d, \text{ hence } \text{im } G_S = K^{2d}, \\ N/2 < E \leq 3N/4 : & \quad \text{rank } G_S = 2d - 1, \text{ and } \text{im } G_E \subseteq \text{im } G_S. \end{aligned}$$

The Shplonk and KZG messages are linear functions of these Gemini coordinates and of the sumcheck value. The sumcheck value is evaluated from the same mask polynomial after the Gemini coordinates are fixed. Hence the sparse mask induces the same distribution on every full-transcript mask direction exposed by dense degree- $< E$ masking.

Proof

1. Reduction to the Gemini block

Let T_E be the linear map from a mask polynomial to the mask-dependent part of the algebraic transcript. In the algebraic group model for KZG, a commitment $[P]$ is represented by the scalar $P(\tau)$. For the KZG path considered here, the mask-dependent transcript coordinates are

$$M(u), \quad M_t(\tau), \quad M_t(-r_t) \quad (0 \leq t < d), \quad Q_M(\tau), \quad W_M(\tau).$$

Here M_t is the t -fold Gemini fold of M , and Q_M, W_M denote the mask contributions to the Shplonk quotient commitment and the KZG witness commitment.

Lemma 1.1 (Shplonk/KZG rows). After the Fiat-Shamir challenges are fixed and the Shplonk/KZG denominators are nonzero, the coordinates $Q_M(\tau)$ and $W_M(\tau)$ are K -linear combinations of

$$M(u), \quad M_t(\tau), \quad M_t(-r_t) \quad (0 \leq t < d).$$

Proof. Shplonk constructs

$$Q(X) = \sum_i \nu^i \frac{P_i(X) - v_i}{X - z_i}.$$

Taking the mask-dependent part and evaluating at $X = \tau$ gives a linear combination of the corresponding mask evaluations, with coefficients determined by ν, τ, z_i and the nonzero denominators $\tau - z_i$. In the Gemini instance, those mask evaluations are exactly the displayed Gemini openings and the sumcheck value. The KZG witness coordinate is obtained from the KZG quotient for opening Q at z ; after substituting the Shplonk expression, its mask-dependent numerator is again a linear combination of the same coordinates, with additional denominators collected in $D_{\text{Shplonk/KZG}}$. $\square$

Thus the only possible independent mask leakage is contained in the augmented Gemini map

$$\widehat{G}_E(P) = (P(u), G_E(P)).$$

The determinant argument below proves the image statement for G_E . The sumcheck coordinate is not prescribed independently by the simulator: once a sparse mask $M \in V_S$ has been chosen to realise the required Gemini leakage, $M(u)$ is the value of that same polynomial. Hence the full transcript is sampled from the pushforward of the sparse coefficient distribution under $\widehat{G}_S$, and Lemma 1.1 shows that Shplonk and KZG add no further rank condition beyond this augmented Gemini data.

For the rank witnesses below we use the $2d \times |S|$ Gemini matrix B , whose rows are

$$M_t(\tau), \quad M_t(-r_t) \quad (0 \leq t < d),$$

and whose columns are the monomials $E_s = X^s$ for $s \in S$.

2. Fold formulas and row normalisation

For $s < N$, set

$$q_t(s) = \left\lfloor \frac{s}{2^t} \right\rfloor, \quad \ell_t(s) = L_{s \bmod 2^t}(u_0, \dots, u_{t-1}).$$

The Gemini fold recursion gives, for the monomial $E_s = X^s$,

$$\text{fold}_t(E_s) = \ell_t(s)E_{q_t(s)}, \quad M_t(x)(E_s) = \ell_t(s)x^{q_t(s)}.$$

Replace each row pair $(M_t(\tau), M_t(-r_t))$ by

$$D_t = M_t(\tau) - M_t(-r_t), \quad M_t = M_t(-r_t).$$

This row operation has determinant 1. Define

$$\phi_m(\tau, y) = \frac{\tau^m - y^m}{\tau - y} = \sum_{i=0}^{m-1} \tau^{m-1-i} y^i, \quad \phi_0 = 0.$$

We use the following elementary identities. For $a \geq b \geq 1$,

$$\phi_a(\tau, y)\phi_{b-1}(\tau, y) - \phi_b(\tau, y)\phi_{a-1}(\tau, y) = -(\tau y)^{b-1}\phi_{a-b}(\tau, y).$$

Indeed, multiply by $(\tau - y)^2$ and use $(\tau - y)\phi_m = \tau^m - y^m$; the mixed terms cancel and leave $-(\tau y)^{b-1}(\tau^{a-b} - y^{a-b})$. We also use the recursions

$$\phi_m(\tau, y) = \tau\phi_{m-1}(\tau, y) + y^{m-1}, \quad \phi_m(\tau, y) = y\phi_{m-1}(\tau, y) + \tau^{m-1}.$$

Then

$$D_t(E_s) = (\tau + r_t)D'_t(E_s), \quad D'_t(E_s) = \ell_t(s)\phi_{q_t(s)}(\tau, -r_t).$$

After extracting the factor $\tau + r_t$ from the row D_t , set

$$M_t^{\text{new}} = M_t + r_t D'_t.$$

Lemma 2.1 (normalised row values). For every t and s ,

$$M_t^{\text{new}}(E_s) = \begin{cases} \ell_t(s), & q_t(s) = 0, \\ \ell_t(s)\tau r_t \phi_{q_t(s)-1}(\tau, -r_t), & q_t(s) \geq 1. \end{cases}$$

Proof. If $q_t(s) = 0$, then $M_t(E_s) = \ell_t(s)$ and $D'_t(E_s) = 0$. If $q_t(s) = m \geq 1$, use

$$\phi_m(\tau, y) = \tau\phi_{m-1}(\tau, y) + y^{m-1}$$

with $y = -r_t$. Then

$$(-r_t)^m + r_t\phi_m(\tau, -r_t) = \tau r_t \phi_{m-1}(\tau, -r_t),$$

and multiplication by $\ell_t(s)$ gives the formula. $\square$

All row operations in this section are invertible over K away from the factors $\tau + r_t$, already included in $\mathcal{E}_E$.

3. Reducing to Block-Triangular Form

The following two identities are the filtration step.

Lemma 3.1 (old-column leakage). If $j < k$ and $s \in C_j$, then

$$M_k^{\text{new}}(E_s) = \prod_{i=j+1}^{k-1} (1 - u_i) M_{j+1}^{\text{new}}(E_s).$$

Proof. For $s \in C_j$, one has $q_k(s) = q_{j+1}(s) = 0$. Lemma 2.1 therefore reduces both sides to Lagrange factors. The bits $j+1, \dots, k-1$ of s are all zero, so the multilinear Lagrange recursion gives

$$\ell_k(s) = \ell_{j+1}(s) \prod_{i=j+1}^{k-1} (1 - u_i).$$

This is the desired identity. $\square$

Lemma 3.2 (adjacent block). If $k \geq 1$ and $s \in C_{k-1}$, then

$$M_k^{\text{new}}(E_s) = u_{k-1} D'_{k-1}(E_s).$$

Proof. For $s \in C_{k-1}$,

$$q_{k-1}(s) = 1, \quad q_k(s) = 0, \quad \text{bit}_{k-1}(s) = 1.$$

Thus $D'_{k-1}(E_s) = \ell_{k-1}(s) \phi_1(\tau, -r_{k-1}) = \ell_{k-1}(s)$, while Lemma 2.1 gives $M_k^{\text{new}}(E_s) = \ell_k(s) = u_{k-1} \ell_{k-1}(s)$. $\square$

For $k \geq 1$, define

$$N_k = M_k^{\text{new}} - u_{k-1} D'_{k-1} - (1 - u_{k-1}) M_{k-1}^{\text{new}}.$$

The transformation from the rows $(D'_0, M_0^{\text{new}}, \dots, D'_{d-1}, M_{d-1}^{\text{new}})$ to

$$(D'_0, M_0^{\text{new}}, D'_1, N_1, \dots, D'_{d-1}, N_{d-1})$$

is unit-triangular in the row index. Lemmas 3.1 and 3.2 imply

$$D'_k|_{C_0 \sqcup \dots \sqcup C_{k-1}} = 0, \quad N_k|_{C_0 \sqcup \dots \sqcup C_{k-1}} = 0.$$

Therefore, with columns ordered $C_0|C_1|\dots|C_{d-1}$, the transformed matrix is block-lower-triangular. Its diagonal blocks are

$$T_k = \begin{pmatrix} D'_k(E_s) \\ N_k(E_s) \end{pmatrix}_{s \in C_k}.$$

The block T_0 has size 2×1 , the middle blocks T_k for $1 \leq k \leq d-2$ have size 2×2 , and the boundary block T_{d-1} has size 2×3 .

4. Middle determinants

For $1 \leq k \leq d-2$, put $m = k-1$. The middle block has columns

$$C_k = \{2^{k+1} - 1, 2^k\}.$$

Define

$$A_m^+(r_m) = u_m + (1 - u_m)r_m, \quad A_m^-(\tau) = u_m - (1 - u_m)\tau.$$

Lemma 4.1 (middle block determinant). For every $1 \leq k \leq d-2$,

$$\det T_k = -L_0(u_{<m})L_{2^m-1}(u_{<m})(\tau - r_m)A_m^+(r_m)A_m^-(\tau).$$

Proof. Let

$$u = u_m, \quad r = r_m, \quad L_- = L_0(u_{<m}), \quad L_+ = L_{2^m-1}(u_{<m}).$$

On the two columns of C_k one has $q_k = 1$. Hence $M_k^{\text{new}} = 0$ and $D'_k = \ell_k$ there, so the D'_k row is

$$(uL_+, (1-u)L_-).$$

Since $M_k^{\text{new}} = 0$ on C_k ,

$$N_k = -uD'_m - (1-u)M_m^{\text{new}}$$

on this block. The level- m quotients of the two columns $2^{k+1} - 1$ and 2^k are 3 and 2. Using

$$\phi_2(\tau, -r) = \tau - r, \quad \phi_3(\tau, -r) = \tau^2 - \tau r + r^2,$$

Lemma 2.1 gives

$$N_k(2^{k+1} - 1) = -L_+(u(\tau^2 - \tau r + r^2) + (1-u)\tau r(\tau - r)),$$

$$N_k(2^k) = -L_-(u(\tau - r) + (1-u)\tau r).$$

Therefore

$$\begin{aligned} \det T_k &= L_+L_- \left((1-u)(u(\tau^2 - \tau r + r^2) + (1-u)\tau r(\tau - r)) \right. \\ &\quad \left. - u(u(\tau - r) + (1-u)\tau r) \right) \\ &= -L_+L_-(\tau - r)(u + (1-u)r)(u - (1-u)\tau). \end{aligned}$$

Substituting back $m = k-1$ gives the claimed formula. The computation depends only on the fixed middle columns $2^{k+1} - 1$ and 2^k , not on the top pair $E-1, E-2$. $\square$

5. Boundary preparation

The first fresh block is $C_0 = \{1\}$. Since

$$D'_0(E_1) = \phi_1(\tau, -r_0) = 1$$

and every row except D'_0 vanishes on C_0 after the block-triangular reduction, the Schur complement at this pivot deletes the row D'_0 and the column 1.

The row M_0^{new} remains unpaired. For each middle block C_k , $1 \leq k \leq d-2$, Lemma 4.1 and the condition $R_E \neq 0$ imply that T_k is invertible. Therefore there are unique $\alpha_k, \beta_k \in K$ such that

$$M_0^{\text{new}}|_{C_k} = \alpha_k D'_k|_{C_k} + \beta_k N_k|_{C_k}.$$

Replace M_0^{new} successively by

$$M_0^{\text{new}} - \alpha_k D'_k - \beta_k N_k \quad (1 \leq k \leq d-2),$$

in increasing k . Since D'_k and N_k vanish on $C_0 \sqcup \dots \sqcup C_{k-1}$, later eliminations do not change earlier blocks. Let the resulting row be $M_{0,\text{elim}}^{\text{new}}$.

After this operation the determinant, or the relevant reduced minor in the low-tail case, is the product of the middle determinants and a boundary determinant supported on C_{d-1} .

6. The ρ -anti-symmetry

Let

$$c_+ = E - 1, \quad c_- = E - 2, \quad h = 2^{d-2}, \quad \rho = c_+ \bmod h.$$

Since E is even, ρ is odd. Set

$$W_+ = L_\rho(u_{<d-2}), \quad W_- = L_{\rho-1}(u_{<d-2}).$$

Thus $W_+ = u_0 H$ and $W_- = (1 - u_0) H$ for the common higher-bit factor H .

Lemma 6.1 (anti-symmetric vanishing). Let R be one of the rows D'_k , M_k^{new} with $k \geq 1$, or N_k with $k \geq 2$. Then

$$W_- R(c_+) - W_+ R(c_-) = 0.$$

Proof. Fix $k \geq 1$. Since c_+ and c_- are consecutive integers with c_+ odd, their binary expansions differ only in bit 0 below level k . Therefore

$$q_k(c_+) = q_k(c_-),$$

and there is a polynomial factor Y_k independent of the bit-0 choice such that

$$D'_k(c_+) = u_0 H_k Y_k, \quad D'_k(c_-) = (1 - u_0) H_k Y_k,$$

where H_k contains the common bits $1, \dots, k-1$. The same factorisation holds for M_k^{new} , because its value is also $\ell_k(s)$ times a function only of $q_k(s)$. Multiplying by $W_- = (1 - u_0) H$ and $W_+ = u_0 H$ gives equal products.

For N_k with $k \geq 2$, use

$$N_k = M_k^{\text{new}} - u_{k-1}D'_{k-1} - (1 - u_{k-1})M_{k-1}^{\text{new}}.$$

Each row on the right has the same anti-symmetry property just proved, so their linear combination does too. $\square$

The row N_1 is exceptional because it contains level-0 rows. Define

$$X_m = (1 - u_0)\phi_m(\tau, -r_0) - u_0\phi_{m-1}(\tau, -r_0).$$

Lemma 6.2 (Casoratian). For every even $E \geq 4$,

$$X_2X_{E-1} - X_3X_{E-2} = \tau r_0\phi_{E-4}(\tau, -r_0)A_0^+(r_0)A_0^-(\tau).$$

Proof. In this proof all ϕ 's are evaluated at $(\tau, -r_0)$. Expanding bilinearly in u_0 and $1 - u_0$ gives

$$\begin{aligned} X_2X_{E-1} - X_3X_{E-2} &= (1 - u_0)^2(\phi_2\phi_{E-1} - \phi_3\phi_{E-2}) \\ &\quad + u_0(1 - u_0)(\phi_3\phi_{E-3} - \phi_{E-1}) \\ &\quad + u_0^2(\phi_{E-2} - \phi_2\phi_{E-3}). \end{aligned}$$

The identity for ϕ gives

$$\phi_2\phi_{E-1} - \phi_3\phi_{E-2} = -\tau^2 r_0^2 \phi_{E-4}, \quad \phi_{E-2} - \phi_2\phi_{E-3} = \tau r_0 \phi_{E-4}.$$

For the middle bracket, use the recursions $\phi_{E-1} = \tau\phi_{E-2} + r_0^{E-2}$ and $\phi_{E-2} = \tau\phi_{E-3} - r_0^{E-3}$, valid because E is even, together with

$$\tau r_0 \phi_{E-5} - \phi_{E-3} = -\phi_2\phi_{E-4},$$

to obtain

$$\phi_3\phi_{E-3} - \phi_{E-1} = -\tau r_0 \phi_2\phi_{E-4}.$$

Substitution yields

$$X_2X_{E-1} - X_3X_{E-2} = \tau r_0 \phi_{E-4} [u_0^2 - u_0(1 - u_0)\phi_2 - (1 - u_0)^2 \tau r_0].$$

Since $\phi_2 = \tau - r_0$, the bracket is

$$(u_0 + (1 - u_0)r_0)(u_0 - (1 - u_0)\tau) = A_0^+(r_0)A_0^-(\tau).$$

This proves the identity. $\square$

Lemma 6.3 (boundary functional). For $d \geq 4$,

$$\Delta_\rho := W_- M_{0,\text{elim}}^{\text{new}}(c_+) - W_+ M_{0,\text{elim}}^{\text{new}}(c_-) = W_+ r_0^2 \tau^2 \frac{\tau^{E-4} - r_0^{E-4}}{\tau^2 - r_0^2}.$$

Proof. Write $W_+ = u_0 H$ and $W_- = (1 - u_0)H$. It suffices to compute the normalised functional

$$\Delta' = (1 - u_0)M_{0,\text{elim}}^{\text{new}}(c_+) - u_0M_{0,\text{elim}}^{\text{new}}(c_-),$$

since $\Delta_\rho = H\Delta'$ and $Hu_0 = W_+$.

Substitute

$$M_{0,\text{elim}}^{\text{new}} = M_0^{\text{new}} - \sum_{k=1}^{d-2} (\alpha_k D'_k + \beta_k N_k).$$

By Lemma 6.1, every eliminated row except N_1 vanishes under Δ' . Thus

$$\Delta' = [(1 - u_0)M_0^{\text{new}}(c_+) - u_0M_0^{\text{new}}(c_-)] - \beta_1[(1 - u_0)N_1(c_+) - u_0N_1(c_-)].$$

Because $M_0^{\text{new}}(E_s) = \tau r_0 \phi_{s-1}(\tau, -r_0)$ for $s \geq 1$,

$$(1 - u_0)M_0^{\text{new}}(c_+) - u_0M_0^{\text{new}}(c_-) = \tau r_0 X_{E-2}.$$

For $N_1 = M_1^{\text{new}} - u_0 D'_0 - (1 - u_0)M_0^{\text{new}}$, the M_1^{new} contribution cancels under Δ' by Lemma 6.1, leaving

$$(1 - u_0)N_1(c_+) - u_0N_1(c_-) = -u_0 X_{E-1} - (1 - u_0)\tau r_0 X_{E-2}.$$

It remains to compute β_1 , the coefficient used to eliminate M_0^{new} on $C_1 = \{3, 2\}$. Cramer's rule against the block T_1 gives

$$\beta_1 = \frac{D'_1(3)M_0^{\text{new}}(2) - D'_1(2)M_0^{\text{new}}(3)}{\det T_1} = \frac{u_0 \tau r_0 - (1 - u_0)\tau r_0 \phi_2}{\det T_1}.$$

Since $u_0 - (1 - u_0)\phi_2 = -X_2$ and Lemma 4.1 gives

$$\det T_1 = -(\tau - r_0)A_0^+(r_0)A_0^-(\tau),$$

we have

$$\beta_1 = \frac{\tau r_0 X_2}{(\tau - r_0)A_0^+(r_0)A_0^-(\tau)}.$$

Let $D_0 = (\tau - r_0)A_0^+(r_0)A_0^-(\tau)$. Combining the preceding displays and multiplying by D_0 gives

$$\Delta' D_0 = \tau r_0 X_{E-2} D_0 + \tau r_0 X_2 [u_0 X_{E-1} + (1 - u_0)\tau r_0 X_{E-2}].$$

A direct expansion from the definitions gives

$$D_0 + (1 - u_0)\tau r_0 X_2 = -u_0 X_3.$$

Therefore

$$\Delta' D_0 = u_0 \tau r_0 (X_2 X_{E-1} - X_3 X_{E-2}).$$

By Lemma 6.2,

$$\Delta' D_0 = u_0 \tau^2 r_0^2 \phi_{E-4}(\tau, -r_0) A_0^+(r_0) A_0^-(\tau).$$

Dividing by D_0 and using $E - 4$ even gives

$$\Delta' = u_0 r_0^2 \tau^2 \frac{\tau^{E-4} - r_0^{E-4}}{\tau^2 - r_0^2}.$$

Multiplying by H yields the claimed formula because $Hu_0 = W_+$. $\square$

7. High-tail rank

Assume

$$E > 3N/4.$$

After the eliminations of Section 5, the remaining boundary columns are $\{c_+, c_-, N/2\}$, and the remaining boundary rows are

$$M_{0,\text{elim}}^{\text{new}}, \quad D'_{d-1}, \quad N_{d-1}.$$

Let U_E^{hi} be this 3×3 boundary matrix.

Lemma 7.1 (high-tail boundary determinant). Up to the fixed sign determined by the column ordering,

$$\det U_E^{\text{hi}} = \pm r_0^2 \tau^2 \frac{\tau^{E-4} - r_0^{E-4}}{\tau^2 - r_0^2} (\tau - r_{d-2}) A_{d-2}^+(r_{d-2}) A_{d-2}^-(\tau) L_0(u_{<d-2}) L_\rho(u_{<d-2}).$$

Proof. Write

$$u = u_{d-2}, \quad r = r_{d-2}, \quad L_0 = L_0(u_{<d-2}).$$

On the boundary columns $c_+, c_-, N/2$, the row D'_{d-1} is

$$(uW_+, uW_-, (1-u)L_0),$$

because $q_{d-1} = 1$ on all three columns. The row N_{d-1} is obtained from

$$N_{d-1} = M_{d-1}^{\text{new}} - uD'_{d-2} - (1-u)M_{d-2}^{\text{new}},$$

and $M_{d-1}^{\text{new}} = 0$ on these columns. For the top pair, $q_{d-2} = 3$; for $N/2$, $q_{d-2} = 2$. Thus

$$N_{d-1}|_{\{c_+, c_-, N/2\}} = (W_+ Y, W_- Y, L_0 Z),$$

where

$$Y = -u\phi_3(\tau, -r) - (1-u)\tau r \phi_2(\tau, -r), \quad Z = -u\phi_2(\tau, -r) - (1-u)\tau r.$$

The cofactor of the $N/2$ entry in the first row is zero, since the two top-pair columns of the lower 2×3 block are proportional to (W_+, W_-) . The other two cofactors are

$$\begin{aligned} C_+ &= uW_-L_0Z - (1-u)L_0W_-Y, \\ C_- &= (1-u)L_0W_+Y - uW_+L_0Z. \end{aligned}$$

Using $\phi_2(\tau, -r) = \tau - r$ and $\phi_3(\tau, -r) = \tau^2 - \tau r + r^2$, the common bracket satisfies

$$uZ - (1-u)Y = -(\tau - r)(u + (1-u)r)(u - (1-u)\tau).$$

Hence the signed cofactor vector is, up to the fixed column-ordering sign,

$$(\tau - r_{d-2})A_{d-2}^+(r_{d-2})A_{d-2}^-(\tau)L_0(u_{<d-2})(W_-, -W_+, 0).$$

Expanding $\det U_E^{\text{hi}}$ along the first row therefore applies the functional $(a, b, c) \mapsto W_-a - W_+b$ to $M_{0,\text{elim}}^{\text{new}}$ on the top pair. Lemma 6.3 evaluates this functional as Δ_ρ , giving the displayed formula. $\square$

The transformed determinant is

$$\det \tilde{B}_E = \left(\prod_{k=1}^{d-2} \det T_k \right) \det U_E^{\text{hi}}.$$

By Lemmas 4.1 and 7.1 this is a nonzero element of K outside $R_E = 0$. Restoring the extracted row factors $\tau + r_t$,

$$\det B_E = \pm r_0^2 \tau^2 (\tau^{E-4} - r_0^{E-4}) \prod_{k=1}^{d-2} (\tau^2 - r_k^2) (\tau + r_{d-1}) \prod_{k=0}^{d-2} A_k^+(r_k) A_k^-(\tau) \mathcal{L}_E^{\text{hi}},$$

where

$$\mathcal{L}_E^{\text{hi}} = \left(\prod_{k=1}^{d-3} L_0(u_{<k}) L_{2^{k-1}}(u_{<k}) \right) L_0(u_{<d-2}) L_\rho(u_{<d-2}).$$

Thus $\det B_E \neq 0$ outside the exceptional locus, so $\text{rank } G_S = 2d$.

8. Low-tail rank and dense containment

Assume

$$N/2 < E \leq 3N/4.$$

Put $m = d - 2$, $u = u_m$, $r = r_m$, and

$$\lambda = -\frac{u(\tau - r) + (1-u)\tau r}{1-u}.$$

Lemma 8.1 (dense hyperplane). For every monomial E_s with $s < 3N/4$, the final triangularised rows satisfy

$$N_{d-1}(E_s) = \lambda D'_{d-1}(E_s).$$

Proof. There are three ranges.

range of s	$q_m(s)$	$\text{bit}_m(s)$	$q_{d-1}(s)$	result
$s < N/4$	0	0	0	both rows vanish
$N/4 \leq s < N/2$	1	1	0	both rows vanish
$N/2 \leq s < 3N/4$	2	0	1	$N_{d-1} = \lambda D'_{d-1}$

Each entry follows by substituting the listed q -values into Lemma 2.1 and the definition of N_{d-1} . $\square$

Therefore $\text{im } G_E$ lies in the hyperplane

$$H_E = \ker(N_{d-1} - \lambda D'_{d-1}).$$

For the sparse support, delete the redundant row N_{d-1} and the boundary column $N/2$. The remaining boundary columns are $\{c_+, c_-\}$, and the remaining boundary rows are $M_{0,\text{elim}}^{\text{new}}$ and D'_{d-1} . On these columns,

$$D'_{d-1} = (1 - u_{d-2})(W_+, W_-).$$

Hence, by Lemma 6.3,

$$\det U_E^{\text{lo}} = (1 - u_{d-2})\Delta_\rho = (1 - u_{d-2})L_\rho(u_{<d-2})r_0^2\tau^2\frac{\tau^{E-4} - r_0^{E-4}}{\tau^2 - r_0^2}.$$

Multiplying by the middle block determinants gives a nonzero $(2d-1) \times (2d-1)$ reduced minor. Thus $\text{rank } G_S = 2d-1$. The functional $N_{d-1} - \lambda D'_{d-1}$ is nonzero as a functional on the transformed row-coordinate space, since the coefficient of the row N_{d-1} is 1. Hence H_E has dimension $2d-1$. Since G_S also satisfies the row relation of Lemma 8.1, its $(2d-1)$ -dimensional image is the full hyperplane H_E . Consequently

$$\text{im } G_E \subseteq H_E = \text{im } G_S.$$

The reduced-minor witness is, up to the fixed ordering sign,

$$\Delta_E^{\text{lo}} = \pm(r_0\tau)^2(\tau - r_0)\frac{\tau^{E-4} - r_0^{E-4}}{\tau^2 - r_0^2} \prod_{k=1}^{d-3}(\tau - r_k) \prod_{k=0}^{d-3} A_k^+(r_k) A_k^-(\tau) \mathcal{L}_E^{\text{lo}},$$

where

$$\mathcal{L}_E^{\text{lo}} = (1 - u_{d-2})L_\rho(u_{<d-2}) \left(\prod_{k=1}^{d-3} L_0(u_{<k}) L_{2k-1}(u_{<k}) \right).$$

For the degenerate padded case $d = 3$, direct symbolic computation gives the same statement with the saturated power $(r_0\tau)^2$ removed; see Appendix B.

9. Specialisation and simulation

The high-tail proof gives $R_E = \det B_E \neq 0$. The low-tail proof gives a nonzero reduced minor $R_E = \Delta_E^{\text{lo}}$. Therefore, over the rational function field K , the claimed rank and image statements hold.

After Fiat-Shamir specialisation over a concrete finite field, failure can occur only when $\mathcal{E}_E = 0$. Schwartz-Zippel gives

$$\Pr[\mathcal{E}_E = 0] \leq \frac{\deg \mathcal{E}_E}{|\mathbb{F}|}$$

for uniformly random independent challenges; in the Fiat-Shamir setting this is the usual algebraic bad-event bound under the random-oracle heuristic.

Outside the exceptional locus, over the concrete field, the coefficient vector $(c_s)_{s \in S}$ is uniform in $\mathbb{F}^{|S|}$, so its image under G_S is uniform on $\text{im } G_S$. In the high-tail case this image is all of $\mathbb{F}^{2d}$. In the low-tail case it is the hyperplane H_E , which contains $\text{im } G_E$. A simulator that knows τ may sample the required point in $\text{im } G_S$, solve a rank-witnessed linear system for sparse coefficients (with one free coefficient in the low-tail case), and then compute all remaining transcript coordinates, including $M(u)$, $Q_M(\tau)$, and $W_M(\tau)$, from that same sparse polynomial. Lemma 1.1 shows that Shplonk and KZG introduce no additional mask direction beyond these computed coordinates.

Appendix A: verification hooks

`SHPLEMINI_ZK_FILTRATION_VERIFY.py` contains exact symbolic and rational checks for the formulas used above. These checks are regression tests for the algebraic identities; the proof obligations are the lemmas stated in the main text.

- `verify_high_tail_rho()` checks the high-tail determinant formula at $d=4$, $E=14,16$, and $d=5$, $E=28,30,32$.
- `verify_low_tail_rho()` checks the low-tail reduced-minor formula at $d=4$, $E=12$, and $d=5$, $E=20,22,24$.
- `verify_boundary_port_decomposition()` divides out the middle block product and checks only the high/low boundary factors. The boundary ratio is 1 in all tested cases.
- `verify_dense_rank_bound()` checks the final-row relation on every monomial at $d=3,4$ and numerically verifies low-tail dense ranks.

The endpoint $E = N$ is included in the same argument: then $\rho = N - 1 \bmod 2^{d-2} = 2^{d-2} - 1$, so the ρ -anti-symmetry reduces to the usual top-pair anti-symmetry.

Appendix B: degenerate $d=3$ low-tail case

For $d=3$, $N=8$, $E=6$, the support is the padded set $S=[5,4,3,2,1,0]$. Delete the redundant row N_2 . All six reduced 5×5 minors factor against

$$G = (\tau - r_0)A_0^+(r_0)A_0^-(\tau) :$$

deleted column s	reduced minor, up to sign
5	$(1-u_0)(1-u_1)G$
4= $N/2$	$u_0(1-u_1)G$
3	$(\tau^2+r_0^2)(1-u_0)(1-u_1)G$
2	$(\tau^2+r_0^2)u_0(1-u_1)G$
1	$r_0^2 \tau^2(1-u_0)(1-u_1)G$
0	$r_0^2 \tau^2 u_0(1-u_1)G$

The canonical deletion of column $N/2=4$ gives the reduced minor required for rank $2d-1=5$.